

Yudi (Mike) Qin

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EDUCATION

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| University of Illinois at Urbana-Champaign
<i>M.S. in Computer Science</i> Thesis Advisor: Nancy M. Amato | Aug. 2022 - Dec. 2024 |
| University of Illinois at Urbana-Champaign
<i>B.S. in Computer Science + Geography & GIS</i> GPA: 3.95/4.0 | Aug. 2018 - May 2022 |

RESEARCH INTERESTS

Robotics, Task and Motion Planning, Multi-Robot Systems, Learning for Planning

PUBLICATIONS

Peer Reviewed Journal

- **M. Qin**, I. Solis, J. D. Motes, M. Morales and N. M. Amato, "K-ARC: Adaptive Robot Coordination for Multi-Robot Kinodynamic Planning," in IEEE Robotics and Automation Letters, vol. 10, no. 10, pp. 10562-10569, Oct. 2025.
- I solis, J. Motes, **M. Qin**, M. Morales and N. M. Amato, "Experience-based Subproblem Planning for Multi-Robot Motion Planning". 2025. In Submission.
- I Solis, J. Motes, **M. Qin**, M. Morales and N. M. Amato, "Adaptive Robot Coordination: A Subproblem-Based Approach for Hybrid Multi-Robot Motion Planning", in IEEE Robotics and Automation Letters, vol. 9, no. 8, pp. 7238-7245, Aug. 2024.

Thesis and Others

- **M. Qin**, I. Solis, J. D. Motes, M. Morales and N. M. Amato, "A Hybrid Framework for Multi-Robot Kinodynamic Planning," Master's Thesis. Champaign, IL. Dec. 2024.

EXPERIENCE

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| Ubtech Robotics Inc.
<i>Robotics Algorithm Engineer, Autonomous Navigation (Full-Time)</i> | Feb. 2025 - Present
Beijing, China |
| <ul style="list-style-type: none">• Designed and implemented a free-space planning module in addition to the existing autonomous navigation stack while regularly maintaining the codebase.• Implemented planning algorithms from state-of-the-art research papers in the freespace-planning module, focusing on improving the overall accuracy and success rate in autonomous parking scenarios.• Developed an Isaac Sim simulation framework alongside the existing Webots simulation framework, which supports the simulation of modules including perception, localization and mapping, planning and control. | |
| Parasol Lab
<i>Graduate Research Assistant</i> | Aug. 2022 - Present
Champaign, IL |
| <ul style="list-style-type: none">• Contributed to the development of the multi-robot system components for PPL(Parasol Planning Library), an open-source C++ robotics task and motion planning library• Developed an algorithm for solving multi-robot motion planning problems, achieving an overall 20% speed-up over existing solutions across various test scenarios.• Extending the algorithm to incorporate trajectory optimizations for real-time controls of the robots. | |
| Foxconn Interconnect Technology
<i>Software Development Engineering Intern</i> | May 2023 - Aug. 2023
Remote |
| <ul style="list-style-type: none">• Developed a software package in ROS for motion planning for multiple AGV(Automated Guided Vehicle) robots.• Worked closely with my mentor to put this package into real usage for the company's factory automation.• Recognized as "Excellent Intern" for consistent contributions and performance during the internship. | |

Parasol Lab

May 2021 - Aug. 2021

Research Experience for Undergraduate (REU) Student

Champaign, IL

- Assisted in developing and implementing multi-robot motion planning algorithms in **C++**.
- Contributed to developing and open-sourcing PPL. Specifically, added testing suites for existing codes, refactored existing codes, and added new components to the library.

CyberGIS Center

Jun. 2020 - Mar. 2022

Student Intern

Champaign, IL

- Developed Python codes for processing large-scale geo-located Twitter data, gathered using Twitter's data collection API.
- Contributed to developing a web-based GIS application that is able to visualize and predict spatiotemporal patterns of infectious diseases.
- Implemented a spatial analytic tool known as MGWR (multi-scale geographically weighted regression) in Python and explored its application on disease data analysis.

TEACHING EXPERIENCE**University of Illinois at Urbana-Champaign**

Aug. 2022 - Present

Graduate Teaching Assistant

Champaign, IL

- CS 441, Applied Machine Learning (Fall 2022)
- CS 421, Programming Languages and Compilers (Spring and Summer 2023)
- CS 440, Intro to Artificial Intelligence (Spring 2024)

HONORS AND AWARDS

UIUC College of Liberal Arts and Science, Dean's List

Fall 2018 - Spring 2022

The Howard and Ruth Roepke Undergraduate Scholarship

Spring 2019